

IIOT Secure Gateway

Secure Prototype with FPGA based Data Diode

TASKS COMPLETED

OpenPLC/SCADA Communication

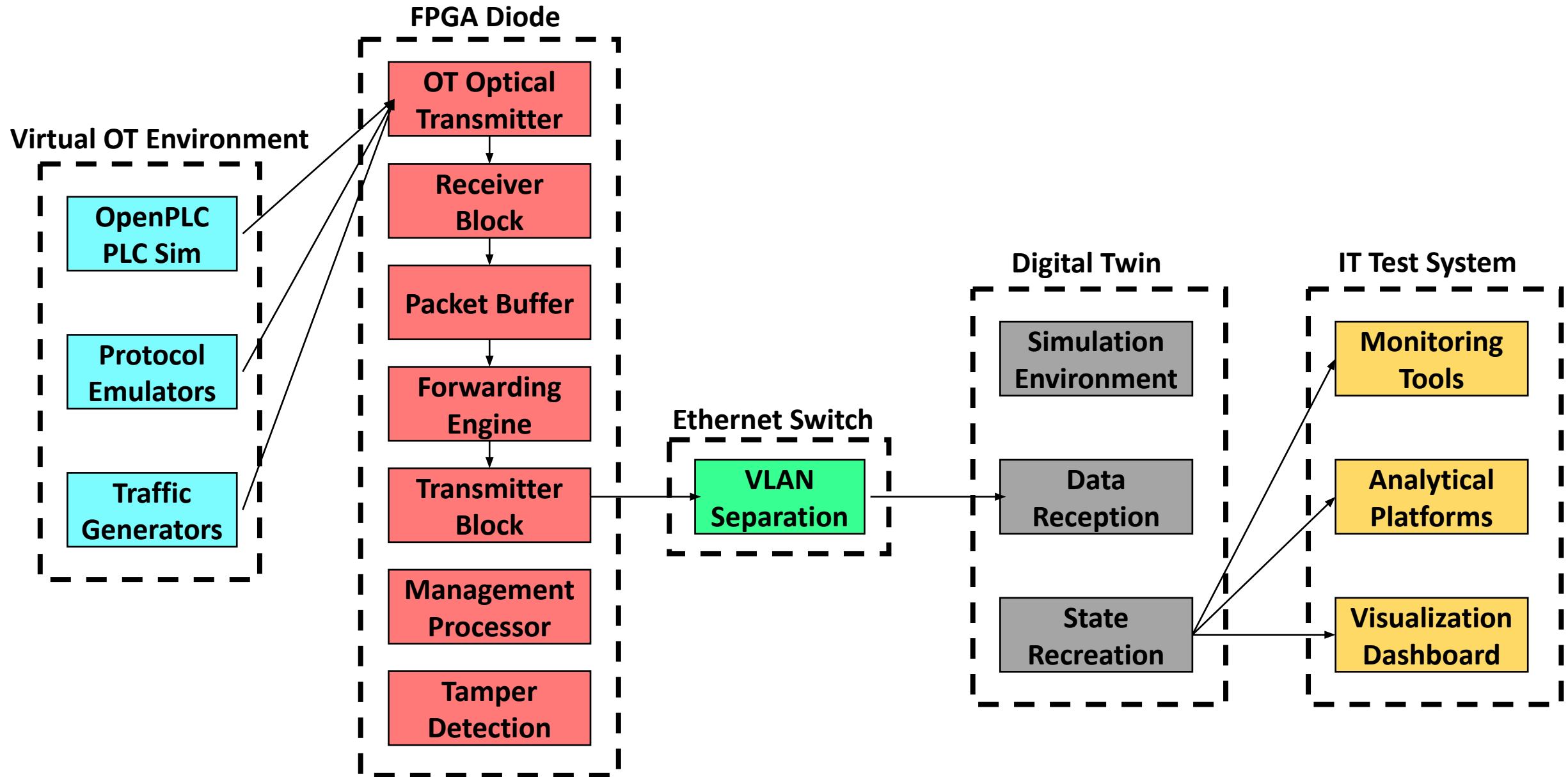
- Installed Docker
- Deployed OpenPLC Runtime in Docker
- Connected ESP with OpenPLC
- Used Modbus TCP for communication
- Integrated SCADA for monitoring and control

FPGA Diode Implementation

- Design of RTL Architecture
- Input FIFO RTL Design
- FIFO Testbench Verification

S no.	Component	Specification	Quantity
1.	FPGA Board	ALINX AX7201 Artix-7 FPGA board	1
2.	SFP Module	Cisco Compatible 1000BASE-SX SFP 850nm 550m DOM Duplex LC/UPC MMF Optical Transceiver Module (Industrial)	4
3.	SFP converter	OneLink Media Converter, Gigabit SFP to RJ45 Fiber Converter, Single Mode SFP Modules, 10/100/1000M RJ45, Auto-Negotiation	2
4.	Fiber optic cables	2m (7ft) Fiber Patch Cable, 1 Fiber, LC UPC Simplex to LC UPC Simplex, Multimode (OM4), LSZH, 1.2mm, Tight-Buffered, Aqua	2
5.	OT Switch	TP-Link 8 Port Gigabit Switch Managed TL-SG108E	1
6.	RJ45 cable	FEDUS Cat6 Ethernet Cable, 10-Meter-High Speed 550MHZ / 10 Gigabit Speed UTP LAN, Network, Internet Cable RJ45 Cable LAN Wire, Patch Computer Cord Gigabit Category 6 Wires for Modem, Router	2
7.	FPGA to PC connector	FEDUS 3 Meter USB 2.0 A to Mini 5 pin B USB Charger Cable	2

DATA DIODE ARCHITECTURE



OPENPLC SETUP

ESP ↔ OpenPLC (Docker) ↔ SCADA

- ESP acts as field device
- OpenPLC acts as PLC
- SCADA used for visualization

Tools and Technologies

1. OpenPLC Runtime
2. Docker
3. ESP32 / ESP8266
4. SCADA Platform
5. Modbus TCP
6. Linux

OPENPLC WEB PLATFORM

NetAccess - IITM Network Po x

10.21.232.151:8080/dashbo x

+

← → ↻

10.21.232.151:8080/dashboard

☆

🔒 👤 📄 ☰

<>

Stopped: led5

OpenPLC User 

 Dashboard

 Programs

 Slave Devices

 Monitoring

 Hardware

 Users

 Settings

 Logout

Status: *Stopped*

Start PLC

Dashboard

Status: **Stopped**

Program: led5

Description:

File: 300230.st

Runtime: N/A

Runtime Logs

I

The screenshot shows the OpenPLC web interface in a browser. The address bar displays the URL: https://10.21.226.8:8006/?console=kvm&novnc=1&vmid=131&vmname=U24-OTenv&node=pve&resize=off. The browser tabs show 'NetAccess - IITM Network Po...', '400 Bad Request', and '10.21.232.151:8080/dashboa...'. The page title is 'Running: led5' and the user is 'OpenPLC User'. The left sidebar contains navigation links: Dashboard (selected), Programs, Slave Devices, Monitoring, Hardware, Users, Settings, and Logout. Below the sidebar, the status is 'Status: Running' and there is a 'Stop PLC' button. The main content area shows the 'Description:' section with 'File: 300230.st' and 'Runtime: 415'. Below this is the 'Runtime Logs' section, which displays a list of error messages: 'Device data_diode is disconnected. Attempting to reconnect...' followed by 'Connection failed on MB device data_diode: Operation now in progress'. This sequence repeats multiple times. At the bottom of the logs, there is a 'Copy logs' button.

Not secure <https://10.21.226.8:8006/?console=kvm&novnc=1&vmid=131&vmname=U24-OTenv&node=pve&resize=off>



NetAccess - IITM Network Po x 400 Bad Request

10.21.232.151:8080/dashboard



🛡️ 🔒 10.21.232.151:8080/dashboard



Running: led5

OpenPLC User



Description:

File: 300230.st

Runtime: 415

Runtime Logs

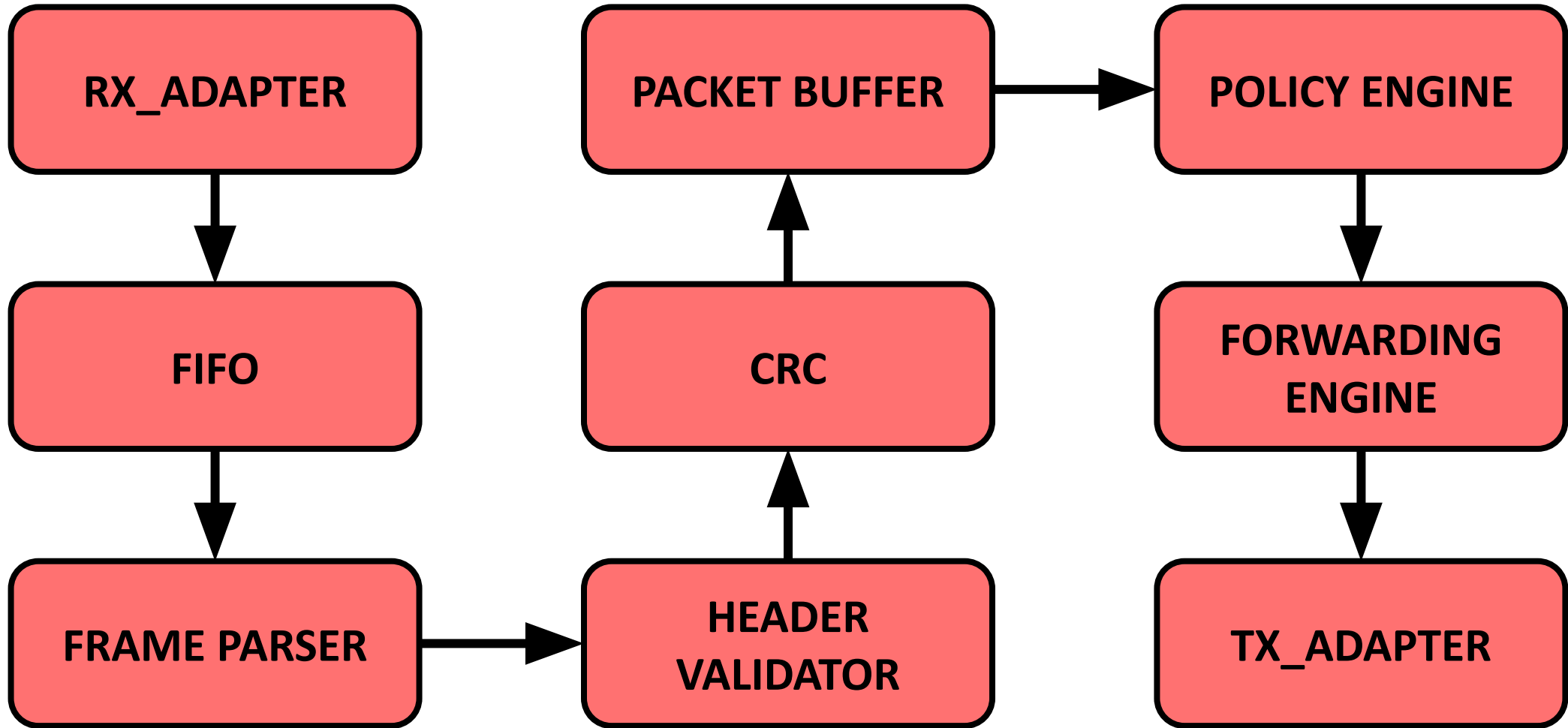
```
Device data_diode is disconnected. Attempting to reconnect...  
Connection failed on MB device data_diode: Operation now in progress  
Device data_diode is disconnected. Attempting to reconnect...  
Connection failed on MB device data_diode: Operation now in progress  
Device data_diode is disconnected. Attempting to reconnect...  
Connection failed on MB device data_diode: Operation now in progress  
Device data_diode is disconnected. Attempting to reconnect...  
Connection failed on MB device data_diode: Operation now in progress  
Device data_diode is disconnected. Attempting to reconnect...  
Connection failed on MB device data_diode: Operation now in progress  
Device data_diode is disconnected. Attempting to reconnect...  
Connection failed on MB device data_diode: Operation now in progress  
Device data_diode is disconnected. Attempting to reconnect...  
Device data_diode is disconnected. Attempting to reconnect...
```

Status: *Running*

Stop PLC

Copy logs

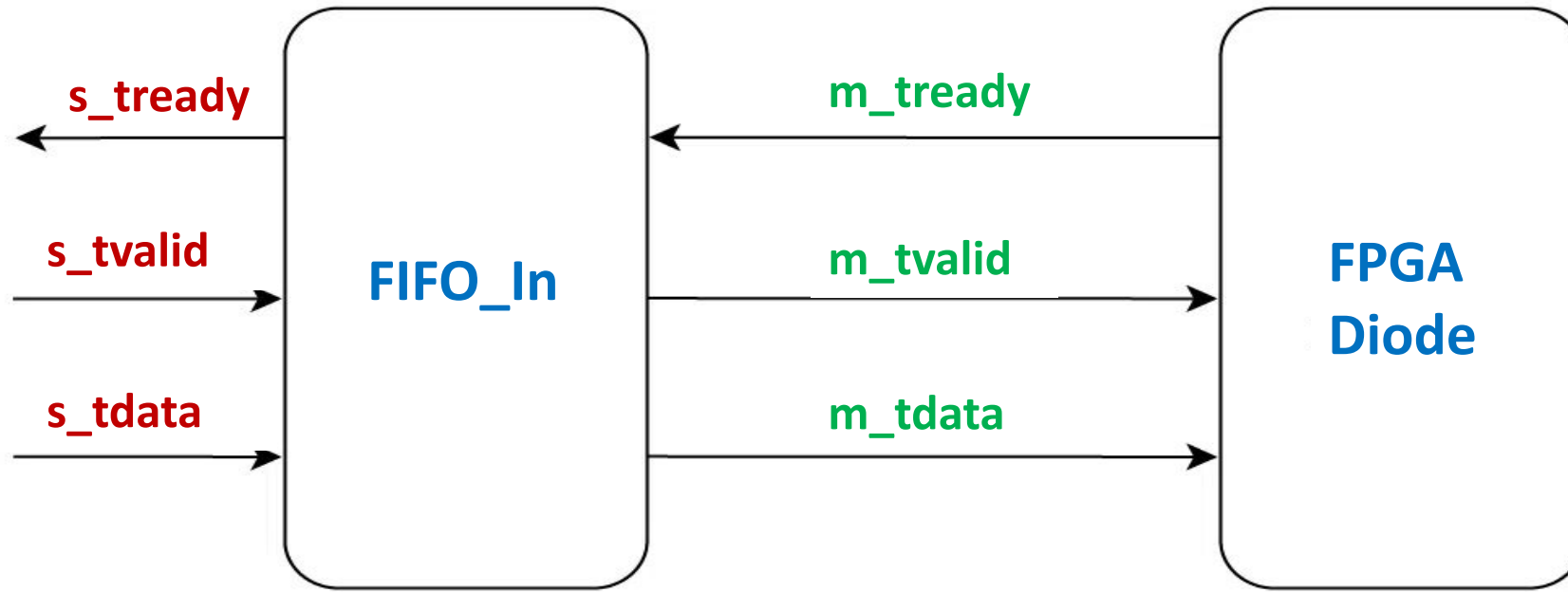
RTL ARCHITECTURE



FIFO DESIGN CHOICES

- Assume **s_tready** = 1: The OT system is always ready to transmit data
- FIFO is **asynchronous** and uses different read and write clocks.
- **No backpressure** allowed: Therefore FIFO cannot be packet atomic.
- Beat Logic is used instead of Packet atomicity.
- Packet corruption is used to filter out partial packets.
- Dropping of overflown/partial packets is preferred to storing them and waiting for ack signal.

PROTOCOL: AXI-STREAM



s_tready Always set to 1 as there should be no backpressure on OT network.

s_tvalid This signals if the FIFO is ready to be written or not

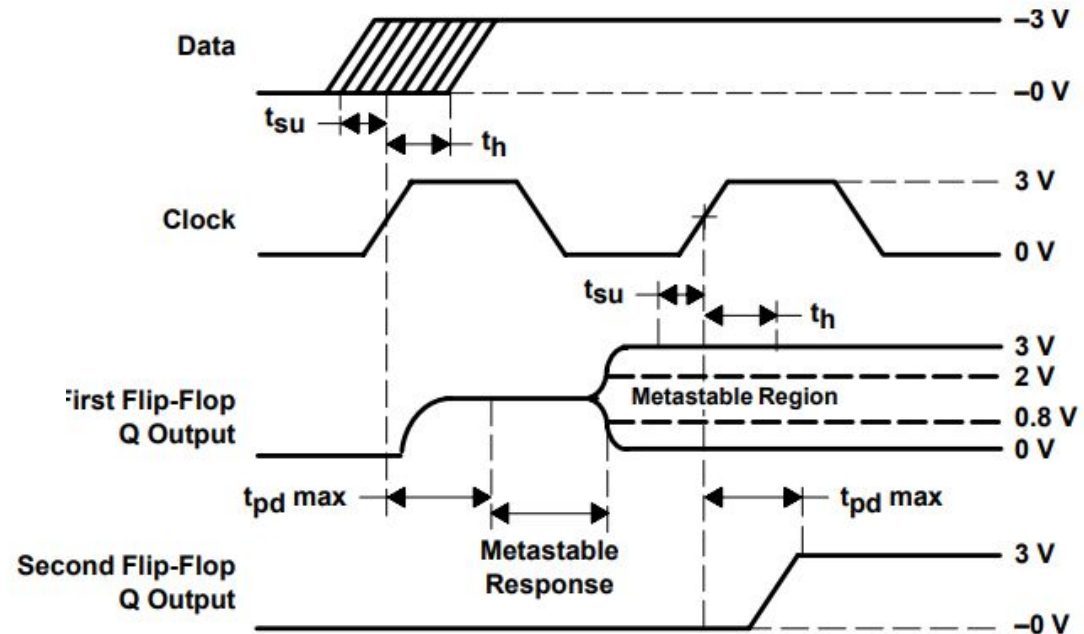
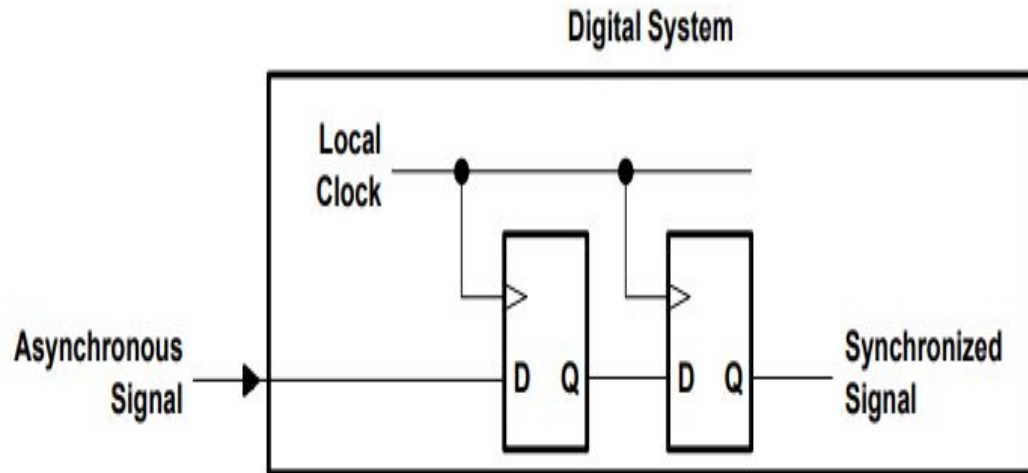
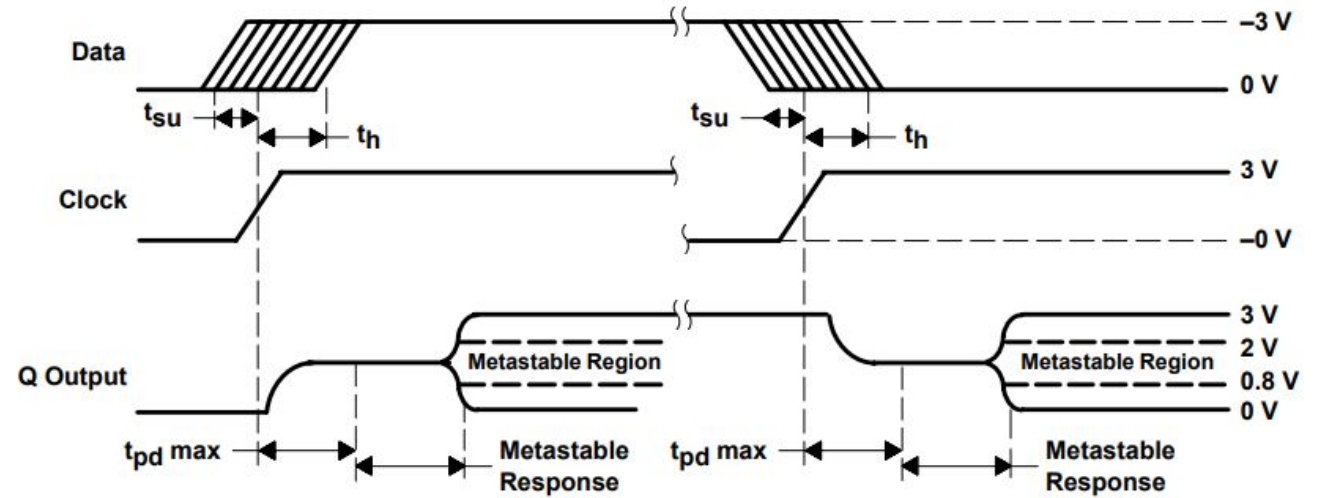
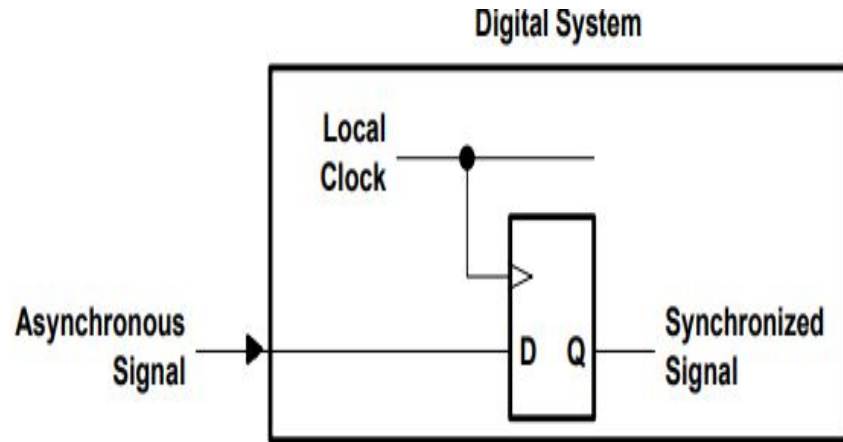
s_tdata Contains DATA USER and LAST bits. It is the data generated from OT side

m_tready Signals if the FPGA diode is ready to receive data

m_tvalid Signals if the FIFO has valid data to be read

m_tdata Data propagated from OT Side to the subsequent stages

CDC MONITORING

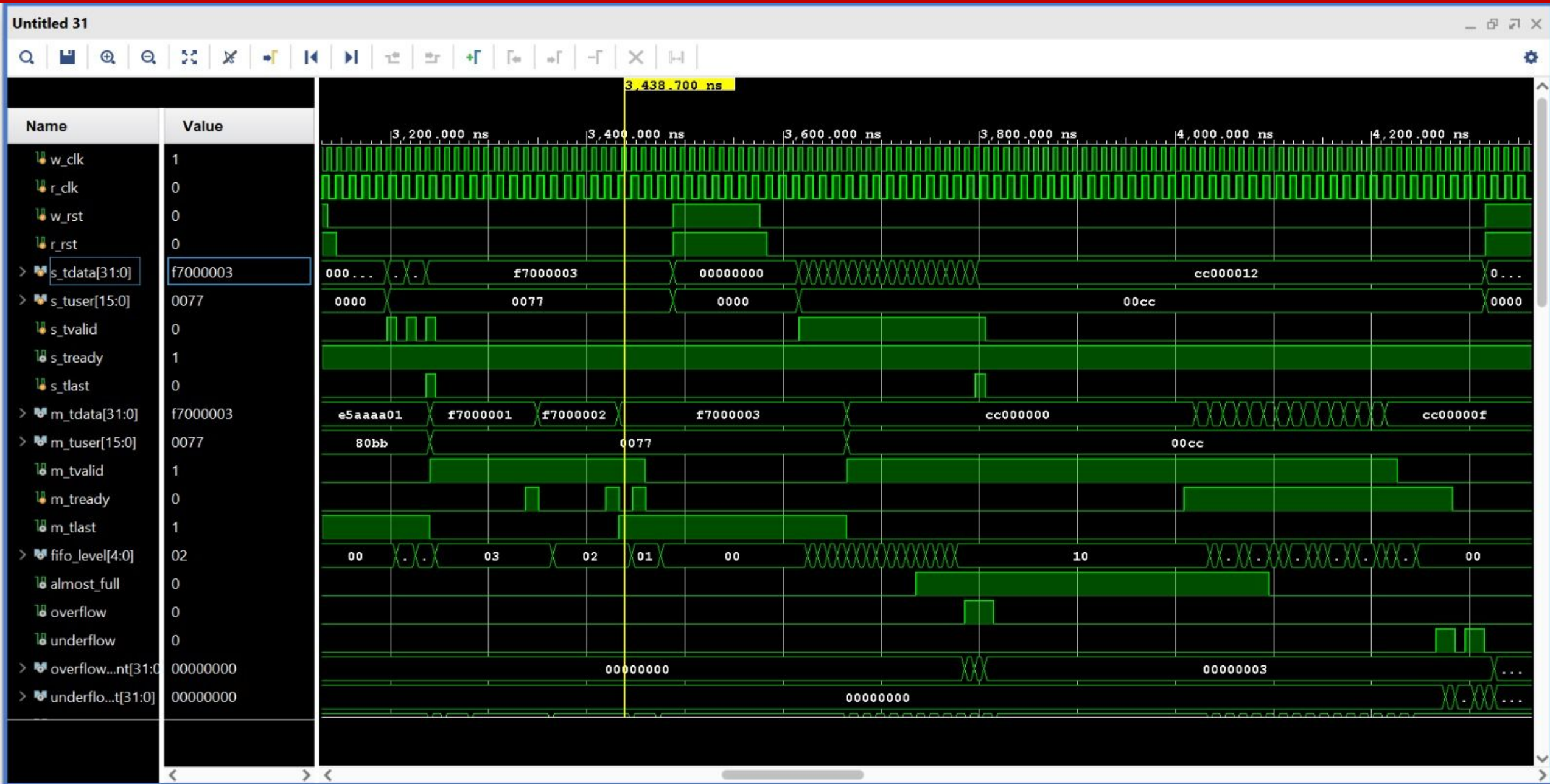


Comparison of 1 vs 2 flops for synchronisation

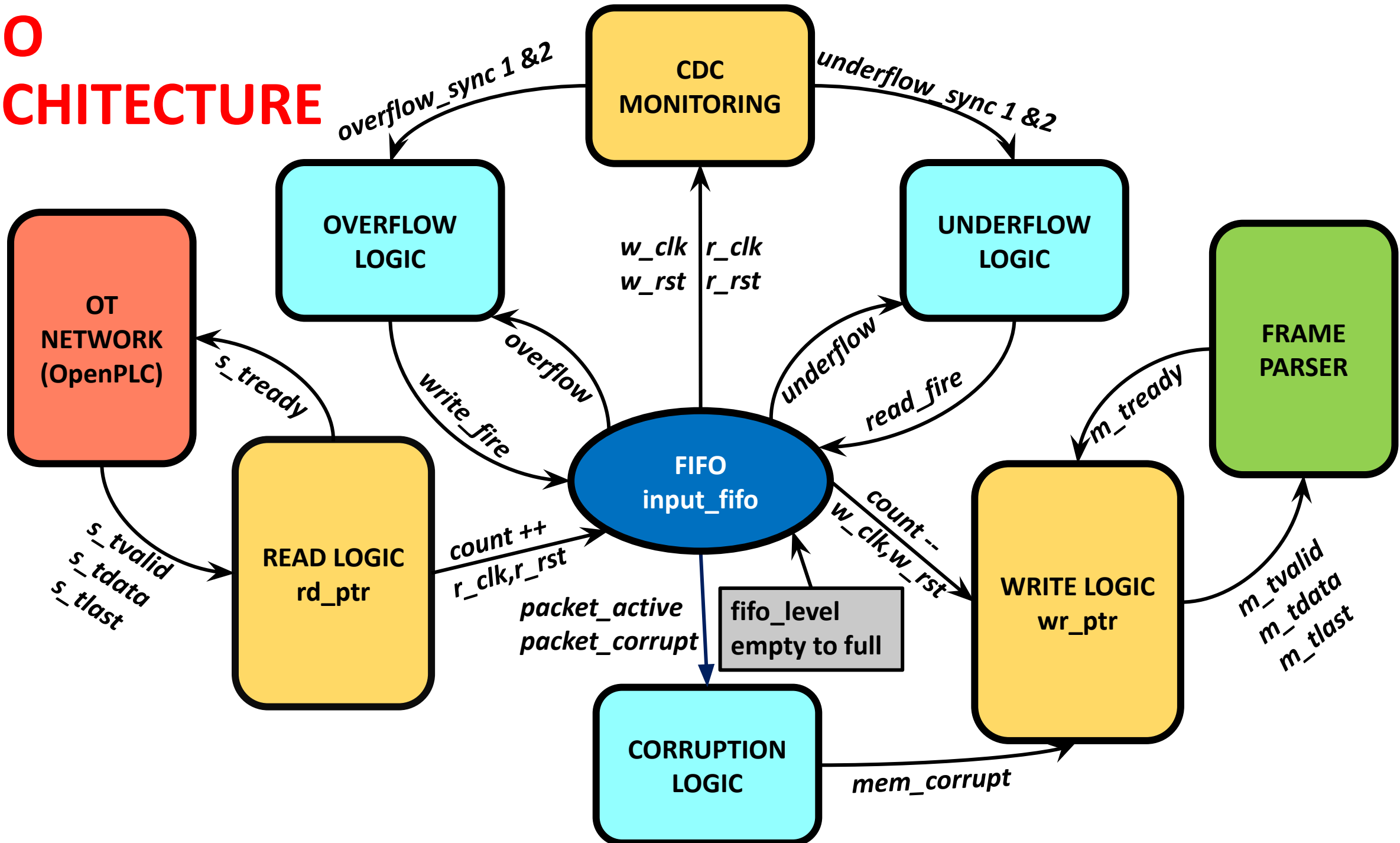
TEST CHECKS

- Test1 — Basic single-beat write/read
- Test2 — Multi-beat packet ordering
- Test3 — Fill to capacity: `fifo_level` and `almost_full`
- Test4 — Overflow / lossy drop
- Test5 — Corrupt packet flagging
- Test6 — Underflow detection
- Test7 — Read-side back-pressure
- Test8 — CDC propagation to monitor domain
- Test9 — `fifo_level_mon` accuracy
- Test10 — Reset mid-operation
- Test11 — Consecutive packets, no corrupt cross-contamination
- Test12 — Stress: random traffic with scoreboard

SIMULATION RESULTS



FIFO ARCHITECTURE



FUTURE WORK

Phase 1 – Virtual OT Deployment

- Develop controlled ladder logic
- Enable Modbus TCP communication
- Generate realistic OT traffic patterns

Phase 2 – Traffic & Protocol Analysis

- Capture and analyze packet size, timing, and frequency
- Study polling intervals and response structure
- Evaluate behavior under constrained (one-way) conditions

Phase 3 – FPGA Integration Testing

- Connect OT simulation to FPGA diode
- Verify strict OT → IT data transfer
- Confirm zero reverse signal path
- Inject malformed and stress traffic

FUTURE WORK

Phase 4 – Performance & Security Validation

- Measure latency introduced by diode
- Evaluate packet drop
- Test continuous load stability
- Validate digital twin state consistency

Final Goal:

Demonstrate a secure, stable, hardware-enforced OT monitoring architecture suitable for industrial deployment.

Long-Term Goal

- Validate FPGA data diode across diverse industrial scenarios
- Enable secure telemetry extraction for monitoring and analytics
- Support predictive maintenance and digital twin environments
- Demonstrate scalability for real industrial deployment

FUTURE SCOPE

- Expanding Beyond Basic Sensor Simulation
- Extend simulation from simple sensor data to full industrial machine environments
- Simulate real-time industrial workflows under unidirectional constraints

Example Use Cases

- CNC Router Machine as a representative OT system
- Real-time motor control signals
- Position feedback loops
- Telemetry extraction without command injection
- Industrial conveyor systems
- Robotic arms and automated assembly lines, Motor-driven manufacturing equipment